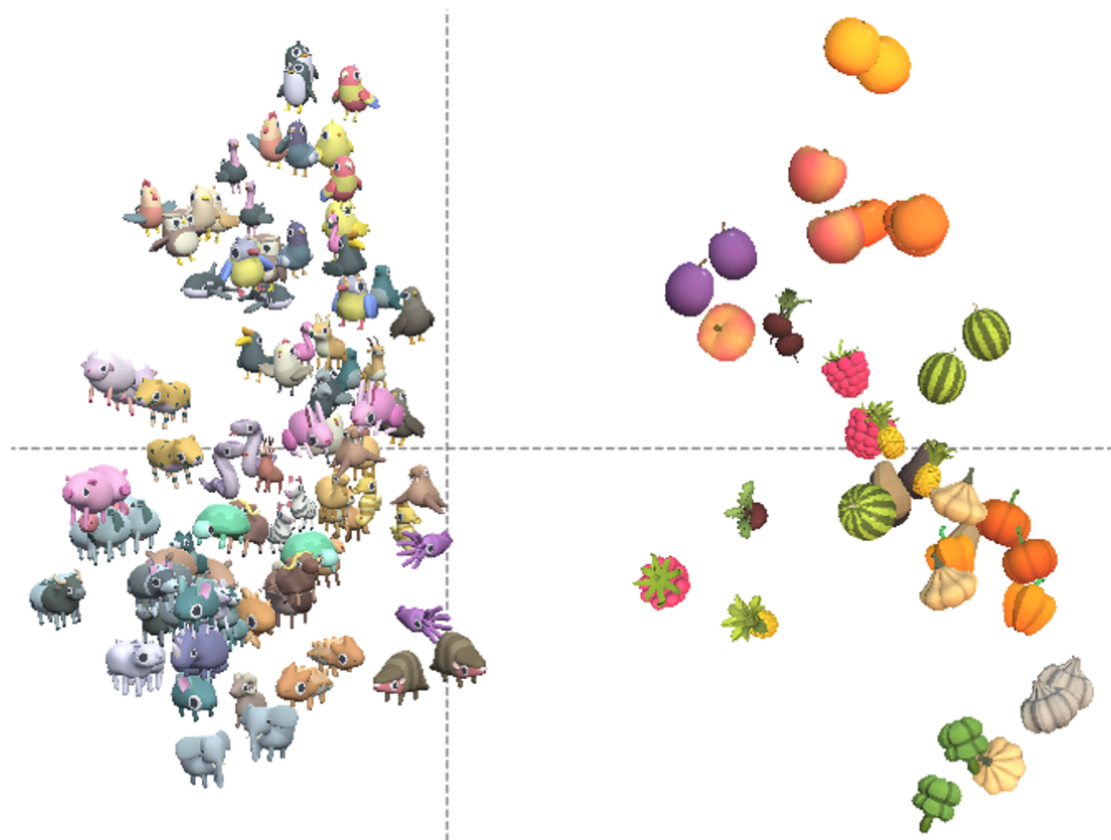
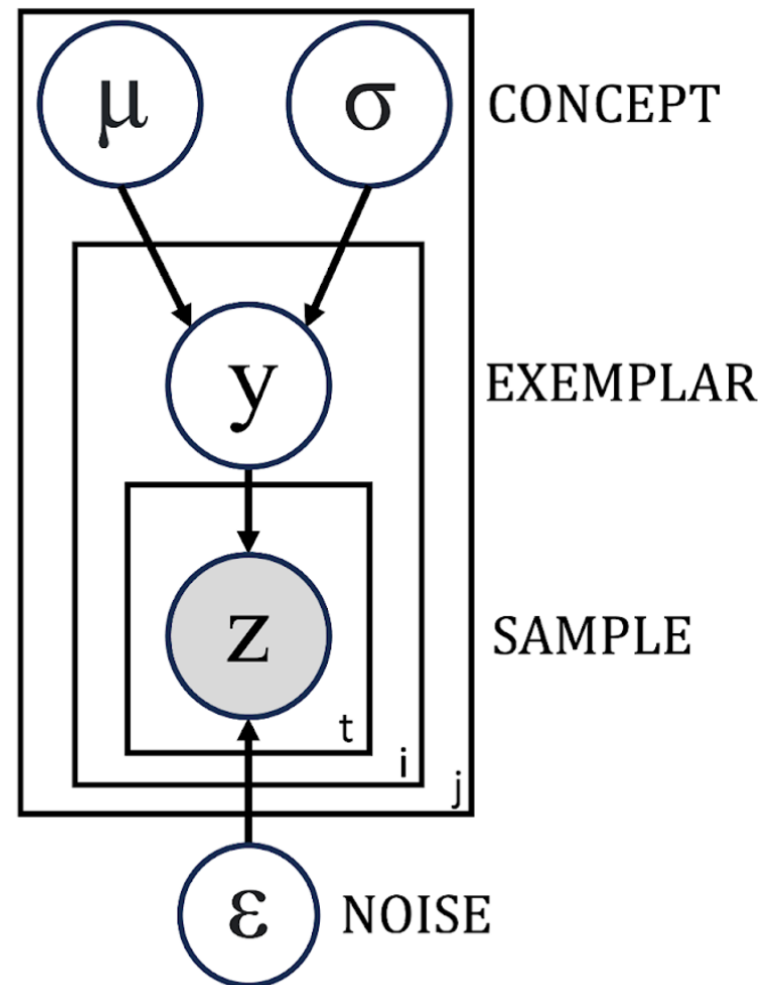


A

Perceptual embeddings

**B**

Learning model

**C**

RANCH sampling algorithm

RANCH algorithm

```

1: for each exemplar  $y$  do
2:    $sample \leftarrow T$ 
3:   while  $sample$  is  $T$ , take another sample  $z$ 
4:     update posterior  $P(\mu, \sigma | z)$ 
5:     compute EIG of next sample  $z_{t+1}$ 
6:     flip  $coin$  with  $p(lookaway) = \frac{EIG(env)}{EIG(z_{t+1}) + EIG(env)}$ 
7:     if  $coin = T$ 
8:        $sample \leftarrow F$ 
9:     end if
10:  end while
11: end for

```
